

“AFT model: log-linear regression to hazard-base formulation”

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AFT model: hazard formulation to log-linear formulation

The AFT model in the log-linear formulation is defined as:

$$\log(T_i) = \beta_0 + x_i^\top \beta + \sigma \epsilon_i,$$

where $\epsilon_i \sim F$ is a fully-specified distribution, and σ is a scale parameter. Then,

$$T_i = \exp\{\beta_0 + x_i^\top \beta\} \exp\{\sigma \epsilon_i\}.$$

Consequently, the survival function is:

$$\begin{aligned} S(t \mid x_i) &= \Pr(T_i > t) \\ &= \Pr(\exp\{\beta_0 + x_i^\top \beta\} \exp\{\sigma \epsilon_i\} > t \mid x_i) \\ &= \Pr\left(\epsilon_i > \frac{\log(t) - \beta_0 - x_i^\top \beta}{\sigma} \mid x_i\right) \\ &= 1 - F\left(\frac{\log(t) - \beta_0 - x_i^\top \beta}{\sigma}\right). \end{aligned}$$

Then, the density function is

$$f(t \mid x_i) = \frac{1}{\sigma t} f\left(\frac{\log(t) - \beta_0 - x_i^\top \beta}{\sigma}\right).$$

Therefore, the hazard function is:

$$h(t \mid x_i) = \frac{f(t \mid x_i)}{S(t \mid x_i)} = \frac{\frac{1}{\sigma t} f\left(\frac{\log(t) - \beta_0 - x_i^\top \beta}{\sigma}\right)}{1 - F\left(\frac{\log(t) - \beta_0 - x_i^\top \beta}{\sigma}\right)}.$$

The baseline hazard represents the model with $x_i = 0$, which is

$$h_0(t) = \frac{\frac{1}{\sigma t} f\left(\frac{\log(t) - \beta_0}{\sigma}\right)}{1 - F\left(\frac{\log(t) - \beta_0}{\sigma}\right)}.$$

Then, we get the hazard-base formulation (we need to multiply by $\exp\{-x_i^\top \beta\}$ to compensate for the same term in the denominator):

$$h(t \mid x_i) = h_0(t \exp\{-x_i^\top \beta\}) \exp\{-x_i^\top \beta\}.$$

By reparameterising $\theta = -\beta$ we obtain the classical expression:

$$h(t \mid x_i) = h_0(t \exp\{x_i^\top \theta\}) \exp\{x_i^\top \theta\}.$$

AFT model: interpretation of regression coefficients.

The corresponding cumulative hazard function is:

$$H(t \mid x_i) = H_0(t \exp\{x_i^\top \theta\}).$$

Consequently, the corresponding survival function is:

$$\begin{aligned} S(t \mid x_i) &= \exp[-H_0(t \exp\{x_i^\top \theta\})] \\ &= S_0(t \exp\{x_i^\top \theta\}). \end{aligned}$$